

Thermonuclear Reactions

Note for NS 3035

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The best learning often begins just beyond our comfort zone

It means that, truly understanding this subject is rarely comfortable at the beginning. Derivations may feel messy, and sometimes the mathematics may appear intimidating. But that discomfort is often a sign that real learning is taking place. So as we move deeper into nuclear astrophysics hang on tight and prepare for rough seas ahead. We are leaving the calm harbor and sailing into open waters !!

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Chapter 1

Reaction Rates & Charged Particle Induced Reactions

1.1 Reaction Rates

The reaction Q-value determines the amount of energy released or absorbed during a nuclear reaction. However, in stellar environments the energy released by an individual reaction is only one part of the problem. For many astrophysical applications our primary interest lies with how many nuclear actions occurring within a given volume during a given interval of time. This quantity is known as the **reaction rate**. The reaction rate plays a central role in the study of stellar energy generation nucleosynthesis and the evolution of stars. Reaction rates connect to the main objectives of our course¹

1. Understanding the synthesis of elements in the Universe
2. Understanding the energy produced by stars

The cross-section is defined as the ratio

$$\sigma = \frac{\text{Number of interactions per unit time}}{(\text{Incident particles per unit area per unit time}) (\text{Number of target nuclei})}$$

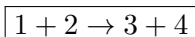
The unit of cross section is area, usually expressed in cm^2 or more conveniently in barns, millibarns, microbarns, and nanobarns. When discussing laboratory measurements we often refer to the *target and projectile*. However inside a stellar plasma there is no distinction between target and projectile because all particles are moving simultaneously. Therefore, the stellar problem must be formulated using the relative motion of the interacting particles. Since our main interest in astrophysics is not only the energy released in a single reaction but also the frequency with which reactions occur, we introduce another concept called *reaction rate*

1.1.1 Reaction Rate Nomenclature

The reaction rate is defined as the number of reactions occurring per unit volume per unit time. It is the most important quantities in Nuclear Astrophysics course because it determines

- the abundance evolution of elements
- the energy production inside stars
- the rate of stellar nucleosynthesis

The reaction rate depends mainly on two quantities: First the reaction cross section and then velocity distribution of particles in the stellar plasma. At finite temperature, particles possess thermal kinetic energy and move in different directions. Consequently, collisions occur continuously, and some of these collisions lead to nuclear reactions. Consider generic nuclear reaction



In laboratory terminology

¹Main objectives of this course is to understand the the origin of the elements in our universe but not as a toddler

- Particle 1 is the projectile
- Particle 2 is the target nucleus
- Particles 3 and 4 are the reaction products

Particle 4 may be a proton, neutron, alpha particle, or gamma ray, while particle 3 is usually the residual nucleus. The reaction cross-section depends on the relative energy of the interacting particles and may therefore be written as $\sigma(E)$, or equivalently $\sigma(v)$. The reaction rate is defined as

$$r_{12} = \frac{N_R}{tV} \quad \text{reaction rate} = \text{reactions per unit volume per unit time}$$

where

- N_R is the number of reactions
- t is the time interval
- V is the interaction volume

The beam current density is defined as

$$j_b = \frac{N_b}{tA}$$

where N_b is the number of beam particles and A is the beam area. Using the definition of cross-section

$$\frac{N_R}{t} = \sigma j_b N_t$$

where N_t is the number of target nuclei. Substituting the expression for j_b

$$\frac{N_R}{t} = \sigma \frac{N_b}{tA} N_t$$

Dividing by the volume V

$$r_{12} = \frac{N_R}{tV} = \sigma \frac{N_t}{V} \frac{N_b}{V} v$$

Defining the number densities

$$N_1 = \frac{N_t}{V} \quad N_2 = \frac{N_b}{V}$$

the reaction rate becomes

$$\boxed{r_{12} = N_1 N_2 v \sigma(v)}$$

where N_1 and N_2 are the number densities of the interacting particles.

1.1.2 Velocity Distribution

At thermodynamic equilibrium, all particles do not move with the same velocity. Instead the relative velocities are described by a probability distribution $P(v)$. The quantity $P(v) dv$ represents the probability that the relative velocity lies between v and $v + dv$. The normalization condition is

$$\int_0^\infty P(v) dv = 1$$

1.1.3 Reaction Rate for a Velocity Distribution

For a distribution of velocities, the reaction rate becomes

$$r_{12} = N_1 N_2 \int_0^\infty v P(v) \sigma(v) dv$$

The integral is defined as

$$\langle \sigma v \rangle_{12} = \int_0^\infty v P(v) \sigma(v) dv$$

and is called the reaction rate per particle pair. Hence,

$$r_{12} = N_1 N_2 \langle \sigma v \rangle_{12}$$

For identical particles,

$$r_{12} = \frac{N_1 N_2 \langle \sigma v \rangle_{12}}{1 + \delta_{12}}$$

where δ_{12} is the Kronecker delta.

1.2 Charged Particle Induced Reactions

In stars, nuclear charge particle induce reactions are initiated by thermal motion. As gravitational contraction increases the temperature, nuclei acquire kinetic energy and collide with one another. Since these reactions originate from thermal motion, they are called thermonuclear reactions

1.3 Maxwell–Boltzmann Distribution

In most stellar environments the velocity distribution is described by the Maxwell Boltzmann distribution

$$P(v) dv = 4\pi v^2 \left(\frac{m_{12}}{2\pi kT} \right)^{3/2} e^{-m_{12}v^2/(2kT)} dv$$

The reduced mass is $m_{12} = \frac{m_1 m_2}{m_1 + m_2}$. The kinetic energy is $E = \frac{m_{12}v^2}{2}$. Differentiating

$$\frac{dE}{dv} = m_{12}v$$

The energy distribution becomes

$$P(E) dE = \frac{2}{\sqrt{\pi}} \frac{\sqrt{E}}{(kT)^{3/2}} e^{-E/kT} dE$$

The most probable velocity is

$$v_T = \sqrt{\frac{2kT}{m_{12}}}$$

and the corresponding energy is

$$E_T = \frac{kT}{2}$$

1.3.1 Reaction Rate Per Charge Particle Pair

Using the Maxwell–Boltzmann distribution, the reaction rate per particle pair is

$$\langle \sigma v \rangle_{12} = \sqrt{\frac{8}{\pi m_{12}}} \frac{1}{(kT)^{3/2}} \int_0^\infty E \sigma(E) e^{-E/kT} dE$$

In practical applications, the quantity tabulated is

$$N_A \langle \sigma v \rangle$$

where N_A is Avogadro's number. The units are

$$\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}$$

Chapter 2

Neutron-Induced Reactions

The nuclear reaction rate was expressed in terms of the reaction cross-section and the velocity distribution of the interacting particles. For charged-particle reactions (In Chapter 1), the velocity distribution of the particles was assumed to follow the Maxwell-Boltzmann distribution. Neutron-induced reactions constitute a special class of nuclear reactions. Unlike charged particles, neutrons do not experience Coulomb repulsion and can therefore interact with nuclei even at very low energies. As a result, neutron capture reactions play a dominant role in the synthesis of elements heavier than iron. This chapter discusses the physics of neutron-induced reactions and derives the corresponding reaction-rate expressions used in nuclear astrophysics.

2.1 Importance of Neutron-Induced Reactions

The elemental abundance curve reveals that nuclei heavier than iron are produced primarily through neutron capture processes. The importance of neutrons arises from the absence of electric charge. Since neutrons carry no charge, they do not experience the Coulomb barrier that strongly suppresses charged particle reactions at low energies. Consequently

- neutron capture can occur at relatively low temperatures
- neutron reactions possess comparatively large cross-sections
- neutron capture becomes an efficient mechanism for producing heavy nuclei

Neutron-induced reactions are responsible for two major nucleosynthesis processes: The slow neutron-capture process (*s-process*) and The rapid neutron-capture process (*r-process*). Both processes contribute significantly to the synthesis of nuclei beyond the iron peak

2.2 Neutron-Induced Reactions Nomenclature

A neutron-induced reaction may be represented as

$$n + A \rightarrow B + x$$

where

- n is the incident neutron
- A is the target nucleus
- B is the residual nucleus
- x denotes the emitted particle

Common examples include (n, γ) , (n, p) , (n, α) , (n, n') . Among these neutron-capture reactions (n, γ) are particularly important for stellar nucleosynthesis.

2.3 Reaction Rate for Neutron-Induced Reactions

The general reaction rate for two non-identical particles is

$$r_{12} = N_1 N_2 \langle \sigma v \rangle$$

where N_1 and N_2 are the number densities, σ is the reaction cross-section and v is the relative velocity. For neutron-induced reactions, the reaction rate becomes

$$r_{nA} = N_n N_A \langle \sigma v \rangle$$

where

- N_n is the neutron number density
- N_A is the number density of the target nucleus

The quantity $\langle \sigma v \rangle$ represents the reaction rate per particle pair.

2.4 Thermal Neutrons

In stellar environments, neutrons are usually produced by previous nuclear reactions and rapidly thermalize through collisions with surrounding matter. As a result, the neutron energies are distributed according to a Maxwell–Boltzmann distribution characterized by the local temperature. The thermal velocity is

$$v_T = \sqrt{\frac{2kT}{m_{12}}}$$

where $m_{12} = \frac{m_n m_A}{m_n + m_A}$ is the reduced mass. Since $m_A \gg m_n$ the reduced mass is approximately

$$m_{12} \approx m_n$$

Therefore

$$v_T \approx \sqrt{\frac{2kT}{m_n}}$$

2.5 Maxwellian Averaged Cross Section

Unlike charged particle reactions neutron-induced reactions often occur within a relatively narrow thermal-energy range. Therefore, it is convenient to define the **Maxwellian Averaged Cross Section (MACS)**. The Maxwellian averaged cross-section is defined by

$$\langle \sigma \rangle_T = \frac{\langle \sigma v \rangle}{v_T}$$

where v_T is the thermal velocity. Multiplying by Avogadro's number gives

$$N_A \langle \sigma \rangle_T = \frac{N_A \langle \sigma v \rangle}{v_T}$$

This quantity is extensively tabulated in nuclear astrophysics databases.

2.6 Derivation of the Maxwellian Averaged Cross Section

The reaction rate per particle pair is

$$\langle \sigma v \rangle = \int_0^\infty v P(v) \sigma_n(v) dv$$

where

$$P(v) = \frac{4}{\sqrt{\pi}} \left(\frac{v}{v_T} \right)^2 e^{-(v/v_T)^2} \frac{1}{v_T}$$

is the normalized Maxwell–Boltzmann velocity distribution. Substituting into the reaction-rate expression yields

$$\langle \sigma v \rangle = \frac{4}{\sqrt{\pi}} \int_0^\infty v \sigma_n(v) \left(\frac{v}{v_T} \right)^2 e^{-(v/v_T)^2} \frac{dv}{v_T}$$

Dividing by v_T

$$\langle \sigma \rangle_T = \frac{4}{\sqrt{\pi}} \frac{1}{v_T} \int_0^\infty v \sigma_n(v) \left(\frac{v}{v_T} \right)^2 e^{-(v/v_T)^2} dv$$

Multiplying by Avogadro's number gives

$$N_A \langle \sigma \rangle_T = \frac{4N_A}{\sqrt{\pi} v_T} \int_0^\infty v \sigma_n(v) \left(\frac{v}{v_T} \right)^2 e^{-(v/v_T)^2} dv$$

This expression is the standard reaction-rate formula for neutron-induced reactions

2.7 Physical Interpretation

The Maxwellian averaged cross-section represents the probability that a thermal neutron will induce a reaction when interacting with a target nucleus. The weighting factor

$$\left(\frac{v}{v_T} \right)^2 e^{-(v/v_T)^2}$$

arises from the Maxwell–Boltzmann velocity distribution. Consequently, neutrons with energies close to the thermal energy contribute most strongly to the reaction rate. The MACS therefore provides a direct measure of the effectiveness of a nucleus in capturing thermal neutrons.

2.8 Neutron Capture Nucleosynthesis

Neutron-capture reactions are the primary mechanism responsible for the production of heavy nuclei beyond iron. A neutron capture reaction is written as $A(n, \gamma)B$. After neutron capture, the resulting nucleus may undergo beta decay

$$n \rightarrow p + e^- + \bar{\nu}_e$$

thereby increasing its atomic number. Repeated neutron captures and beta decays produce progressively heavier elements via

1. Slow neutron capture (*s-process*)
2. Rapid neutron capture (*r-process*)

The *s-process* occurs when neutron capture is slower than beta decay whereas the *r-process* occurs when neutron capture is faster than beta decay. Neutron-induced reactions possess several important advantages compared with charged particle reactions

1. No Coulomb barrier
2. Large cross-sections at low energies
3. Efficient production of heavy nuclei
4. Strong sensitivity to neutron density and temperature

2.9 Conclusion

Neutron induced reactions play a central role in nuclear astrophysics particularly in the production of elements heavier than iron. The reaction rate is given by

$$r_{nA} = N_n N_A \langle \sigma v \rangle$$

The thermal velocity is

$$v_T = \sqrt{\frac{2kT}{m_{12}}}$$

The Maxwellian averaged cross-section is defined by

$$\langle\sigma\rangle_T = \frac{\langle\sigma v\rangle}{v_T}$$

The corresponding astrophysical quantity is

$$N_A\langle\sigma\rangle_T$$

Neutron-capture reactions provide the foundation for the s -process and r -process nucleosynthesis pathways and are therefore responsible for the production of most nuclei heavier than iron.

Chapter 3

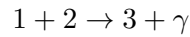
Gamma Induced Reactions

In the previous chapters we discussed charge particle and neutron induced reactions. Neutrons are massive neutral particles and their reaction rates can be treated using a Maxwell-Boltzmann velocity distribution *even though they depends less on MB distribution*. Gamma rays, however are photons. They have no rest mass and no electric charge and therefore their reaction rate must be treated differently. Gamma induced reactions play an important role in hot stellar environments. At sufficiently high temperatures, gamma rays produced in one nuclear reaction can interact with nuclei and induce further reactions. Such reactions become especially important during explosive nucleosynthesis where the photon bath contains high-energy gamma rays

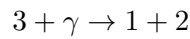
The purpose of this chapter is to derive the reaction rate expression for gamma-induced reactions and to explain its physical meaning

3.1 Gamma Induced Reactions Nomenclature

Consider the radiative capture reaction



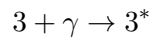
In this reaction, particles 1 and 2 combine to form nucleus 3, and the excess energy is emitted as a gamma ray. These type of nuclear reactions are called **radiative capture reaction**. The inverse of radiative capture is



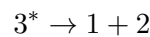
This process is called a **photodisintegration reaction**. In this case, nucleus 3 absorbs a gamma ray and then becomes excited. At last it breaks into particles 1 and 2. Thus $1 + 2 \rightarrow 3 + \gamma$ is a radiative capture reaction, while $3 + \gamma \rightarrow 1 + 2$ is a gamma-induced or photon-induced reaction.

3.2 Physical Picture of Gamma-Induced Reactions

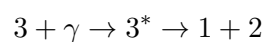
When a gamma ray interacts with a nucleus, it does not change the charge or mass number immediately. Instead, the gamma ray transfers energy to the nucleus and excites it to a higher nuclear energy state. Symbolically,



where 3^* denotes the excited state of nucleus 3. The excited nucleus may then de-excite in different ways. It may emit one or more gamma rays, or if the excitation energy is sufficiently large, it may disintegrate into lighter particles



Therefore, the complete photodisintegration process may be written as



Photodisintegration reactions are usually associated with negative Q -values, because energy must be supplied to break the nucleus apart.

3.3 Reaction Rate for Gamma-Induced Reactions

For an ordinary two-particle reaction, the reaction rate is proportional to the *number densities of the two reacting species*, the *relative velocity*, and the *reaction cross-section*. For a gamma-induced reaction involving nucleus 3 the basic reaction rate may be written as

$$r_{\gamma 3} = N_3 N_\gamma c \sigma(E_\gamma)$$

where

- N_3 is the number density of nucleus 3
- N_γ is the number density of gamma rays
- c is the speed of light
- $\sigma(E_\gamma)$ is the gamma-induced reaction cross-section
- E_γ is the gamma-ray energy

Unlike massive particles, gamma rays always move with speed c . Therefore, the relative velocity factor is replaced by the speed of light. However, in a stellar plasma's gamma rays do not all have the same energy. Their number density depends on energy and temperature. Therefore N_γ must be written as an energy distribution

$$N_\gamma = N_\gamma(E_\gamma)$$

The reaction rate must then be integrated over all possible gamma ray energies

$$r_{\gamma 3} = N_3 \int_0^\infty c N_\gamma(E_\gamma) \sigma(E_\gamma) dE_\gamma$$

This is the general reaction-rate expression for a gamma-induced reaction.

3.4 Gamma-Induced Decay Constant

Since a gamma-induced reaction begins with the absorption of a photon by nucleus 3, it is often useful to express the reaction in terms of a decay constant. The gamma-induced decay constant is defined as the reaction rate per nucleus 3

$$\lambda_\gamma^{(3)} = \frac{r_{\gamma 3}}{N_3}$$

Using the expression for $r_{\gamma 3}$, we obtain

$$\lambda_\gamma^{(3)} = \int_0^\infty c N_\gamma(E_\gamma) \sigma(E_\gamma) dE_\gamma$$

Thus, the decay constant gives the probability per unit time that nucleus 3 absorbs a gamma ray and undergoes a gamma-induced reaction

3.5 Photon Distribution in Stellar Plasma

Charged particles and neutrons in stellar plasma are described approximately by the Maxwell–Boltzmann distribution. Gamma rays, however, are photons and do not obey a Maxwell–Boltzmann distribution. In thermal equilibrium, photons are described by Planck's radiation law. The energy density of photons with frequency between ν and $\nu + d\nu$ is

$$u(\nu)d\nu = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/kT} - 1} d\nu$$

where h is Planck's constant, k is Boltzmann's constant, T is the temperature and c is the speed of light. Using $E_\gamma = h\nu$, we may rewrite Planck's law in terms of the gamma-ray energy. Since

$$dE_\gamma = h d\nu$$

the energy density becomes

$$u(E_\gamma)dE_\gamma = \frac{8\pi}{(hc)^3} \frac{E_\gamma^3}{e^{E_\gamma/kT} - 1} dE_\gamma$$

3.6 Gamma-Ray Number Density

The energy density $u(E_\gamma)dE_\gamma$ gives the energy per unit volume contained in photons with energies between E_γ and $E_\gamma + dE_\gamma$. To obtain the number density of photons, this energy density must be divided by the photon energy E_γ

$$N_\gamma(E_\gamma)dE_\gamma = \frac{u(E_\gamma)dE_\gamma}{E_\gamma}$$

Therefore

$$N_\gamma(E_\gamma)dE_\gamma = \frac{8\pi}{(hc)^3} \frac{E_\gamma^2}{e^{E_\gamma/kT} - 1} dE_\gamma$$

This expression gives the number of gamma rays per unit volume with energies between E_γ and $E_\gamma + dE_\gamma$

3.7 Final Expression for the Gamma-Induced Decay Constant

Substituting the photon number density into the gamma-induced decay constant,

$$\lambda_\gamma^{(3)} = \int_0^\infty c N_\gamma(E_\gamma) \sigma(E_\gamma) dE_\gamma$$

we obtain

$$\lambda_\gamma^{(3)} = \int_0^\infty c \left[\frac{8\pi}{(hc)^3} \frac{E_\gamma^2}{e^{E_\gamma/kT} - 1} \right] \sigma(E_\gamma) dE_\gamma$$

Simplifying

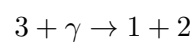
$$\lambda_\gamma^{(3)} = \frac{8\pi}{h^3 c^2} \int_0^\infty \frac{E_\gamma^2}{e^{E_\gamma/kT} - 1} \sigma(E_\gamma) dE_\gamma$$

This is the central result for gamma-induced reactions. It shows that the gamma-induced decay constant depends on

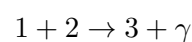
1. the photon energy distribution
2. the gamma-induced reaction cross-section
3. the temperature of the stellar plasma

3.8 Threshold Energy

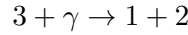
Photodisintegration reactions require the gamma ray to supply enough energy to break the nucleus apart. For the reaction



the gamma-ray energy must exceed the threshold energy. This threshold is related to the Q-value of the inverse reaction. If



has positive Q-value, then the inverse reaction



requires an input energy approximately equal to that Q-value. Thus,

$$E_\gamma \geq E_{\text{thr}}$$

For this reason, the integral for the gamma-induced decay constant is often written as

$$\lambda_\gamma^{(3)} = \frac{8\pi}{h^3 c^2} \int_{E_{\text{thr}}}^{\infty} \frac{E_\gamma^2}{e^{E_\gamma/kT} - 1} \sigma(E_\gamma) dE_\gamma$$

The lower limit is no longer zero, because photons below the threshold energy cannot induce photodisintegration

3.9 Temperature Dependence

The Planck distribution contains the exponential factor

$$e^{E_\gamma/kT} - 1$$

At low temperatures, the number of high-energy photons is very small. Therefore the photodisintegration reactions are strongly suppressed. At high temperatures the photon bath contains more energetic gamma rays. Consequently, gamma-induced reactions become increasingly important. This explains why radiative capture reactions dominate at relatively low temperatures while photodisintegration reactions become significant in hot stellar environments. In explosive nucleosynthesis, such as supernova environments, gamma-induced reactions can strongly affect the final abundance distribution of nuclei.

3.10 Comparison with Charged-Particle and Neutron-Induced Reactions

Gamma-induced reactions differ from charged-particle and neutron-induced reactions in several important ways. For charged particles and neutrons, the reaction rate is written in terms of a Maxwell–Boltzmann velocity distribution

$$\langle \sigma v \rangle = \int_0^\infty v \sigma(v) P(v) dv$$

For gamma rays, the reaction rate is instead written using the photon number density from Planck's law

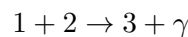
$$\lambda_\gamma^{(3)} = \int_0^\infty c N_\gamma(E_\gamma) \sigma(E_\gamma) dE_\gamma$$

The main differences are

1. Gamma rays move with speed c
2. Gamma rays have no rest mass
3. Gamma rays do not follow the Maxwell–Boltzmann distribution
4. The photon number density is determined by Planck's radiation law
5. The reaction is often expressed through a decay constant λ_γ

3.11 Radiative Capture as the Forward Reaction

A radiative capture reaction is written as



Here two particles combine to form a heavier nucleus, and a gamma ray carries away the excess energy. This process is important because the newly formed nucleus must lose energy in order to remain bound. The gamma ray provides the mechanism by which this energy is removed. Examples of radiative capture reactions include

$$p + A \rightarrow B + \gamma$$

$$n + A \rightarrow B + \gamma$$

$$\alpha + A \rightarrow B + \gamma$$

In compact notation these are written as $A(p, \gamma)B$, $A(n, \gamma)B$, $A(\alpha, \gamma)B$

3.12 Photodisintegration as the Inverse Reaction

The inverse of radiative capture is photodisintegration

$$3 + \gamma \rightarrow 1 + 2$$

In this process, a gamma ray is absorbed by nucleus 3. The nucleus is excited and may disintegrate if the photon energy is sufficiently large. The reaction may be written in compact notation as

$$3(\gamma, 1)2$$

or more generally as

$$A(\gamma, x)B$$

where x may be a neutron, proton, or alpha particle. Common examples include

$$(\gamma, n), \quad (\gamma, p), \quad (\gamma, \alpha)$$

These reactions are especially important in high-temperature environments where energetic gamma rays are abundant.

3.13 Physical Meaning of the Decay Constant

The decay constant $\lambda_\gamma^{(3)}$ has units of inverse time. It measures the probability per unit time that a nucleus 3 will absorb a gamma ray and undergo photodisintegration. If N_3 nuclei are present, the total reaction rate is

$$r_{\gamma 3} = N_3 \lambda_\gamma^{(3)}$$

Thus, the abundance of nucleus 3 changes according to

$$\frac{dN_3}{dt} = -\lambda_\gamma^{(3)} N_3$$

if photodisintegration is the only process destroying nucleus 3. The solution is

$$N_3(t) = N_3(0)e^{-\lambda_\gamma^{(3)}t}$$

Therefore, a larger value of $\lambda_\gamma^{(3)}$ corresponds to faster destruction of nucleus 3 by gamma rays.

3.14 Role in Stellar Nucleosynthesis

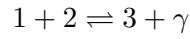
Gamma-induced reactions are especially important in hot environments. At low temperatures, the number of photons with energy above the nuclear threshold is very small. In such environments radiative capture reactions usually dominate. At high temperatures, however, the energetic tail of the photon distribution becomes significant. Photodisintegration reactions can then compete with or even dominate over, radiative capture reactions. This competition between capture and photodisintegration is important in several astrophysical processes, including

- explosive oxygen and silicon burning
- supernova nucleosynthesis
- the p -process
- nuclear statistical equilibrium (NSE)

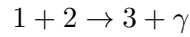
In these environments, gamma-induced reactions can remove neutrons, protons, or alpha particles from nuclei and thereby modify the path of nucleosynthesis.

3.15 Connection to Forward and Inverse Reactions

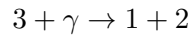
Radiative capture and photodisintegration form a pair of forward and inverse reactions



The forward reaction is



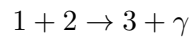
while the inverse reaction is



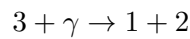
The rates of these two reactions are not independent. In thermodynamic equilibrium, they are related by the principle of detailed balance. This connection is extremely useful because it is often difficult to measure photodisintegration cross-sections directly. Instead, one may measure the radiative capture cross-section and use inverse-reaction theory to determine the photodisintegration rate. The detailed relationship between forward and inverse reaction rates will be developed in the next chapter.

3.16 Summary

Gamma-induced reactions occur when photons interact with nuclei and induce nuclear transformations. A radiative capture reaction is written as



while the inverse photodisintegration reaction is



The gamma-induced reaction rate is

$$r_{\gamma 3} = N_3 \int_0^\infty c N_\gamma(E_\gamma) \sigma(E_\gamma) dE_\gamma$$

The corresponding decay constant is

$$\lambda_\gamma^{(3)} = \frac{r_{\gamma 3}}{N_3}$$

Using Planck's radiation law, the photon number density is

$$N_\gamma(E_\gamma) dE_\gamma = \frac{8\pi}{(hc)^3} \frac{E_\gamma^2}{e^{E_\gamma/kT} - 1} dE_\gamma$$

Therefore

$$\lambda_{\gamma}^{(3)} = \frac{8\pi}{h^3 c^2} \int_{E_{\text{thr}}}^{\infty} \frac{E_{\gamma}^2}{e^{E_{\gamma}/kT} - 1} \sigma(E_{\gamma}) dE_{\gamma}$$

This expression is the fundamental reaction-rate formula for gamma-induced nuclear reactions in stellar environments.

Chapter 4

Inverse Reactions

In the previous chapters, reaction rates were derived for *charged-particle*, *neutron-induced*, and *gamma-induced* nuclear reactions. However, in stellar environments nuclear reactions rarely occur in only one direction. Whenever a reaction can proceed from an initial state to a final state, the reverse process may also occur provided sufficient energy is available. The study of **inverse reactions** is thus one of the most important topics in nuclear astrophysics. Stellar abundances, reaction networks and energy production depend not only on the rate of the forward reaction but also on the rate of its reverse process. This chapter introduces the physical principles of inverse reactions and develops the theoretical framework required to relate forward and inverse reaction cross-sections and reaction rates

4.1 Introduction

Consider a nuclear reaction

$$1 + 2 \rightarrow 3 + 4$$

In this notation, particles 1 and 2 form the entrance channel, while particles 3 and 4 form the exit channel. In laboratory experiments attention is usually focused on the forward reaction. However, inside stars nuclei exist in a hot plasma where particles continuously collide with one another. Consequently the reverse process

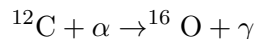
$$3 + 4 \rightarrow 1 + 2$$

may also occur. This reverse process is known as the **inverse reaction**. The existence of inverse reactions is a direct consequence of the laws of energy conservation and quantum mechanics. If a particular transition is allowed in one direction, the reverse transition is generally allowed as well, provided the necessary energy is available. Inverse reactions become especially important in high-temperature astrophysical environments where thermal energies are sufficiently large to overcome reaction thresholds.

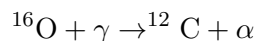
4.2 Forward and Inverse Reactions

A forward reaction may be represented schematically as $1 + 2 \rightarrow 3 + 4$. The corresponding inverse reaction is $3 + 4 \rightarrow 1 + 2$. Both reactions involve exactly the same nuclei and the same nuclear interaction. The only difference is the direction in which the reaction proceeds.

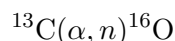
For example



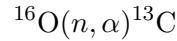
is a radiative capture reaction. Its inverse reaction is



Similarly



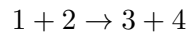
has the inverse reaction



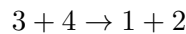
Thus, every reaction possesses a corresponding inverse reaction. The central question is whether the forward and inverse cross-sections are related. The answer is *yes* and the relationship arises from the symmetry properties of quantum mechanics.

4.3 Energetics and the Q-Value

The energetics of a nuclear reaction are described by the reaction Q-value. For the reaction



the Q-value is $Q = (m_1 + m_2 - m_3 - m_4)c^2$. If $Q > 0$, the reaction releases energy and is called an **exothermic reaction**. If $Q < 0$, the reaction absorbs energy and is called an **endothermic reaction**. For the inverse reaction



the Q-value becomes $Q_{\text{inv}} = (m_3 + m_4 - m_1 - m_2)c^2$. Therefore,

$$Q_{\text{inv}} = -Q$$

This is one of the most important properties of inverse reactions. A reaction that releases energy in the forward direction requires exactly the same amount of energy to proceed in the reverse direction. For example, if

$$Q = 5 \text{ MeV}$$

then the inverse reaction requires at least 5 MeV of energy. Consequently, inverse reactions are strongly suppressed at low temperatures and become important only when the thermal energy becomes comparable to the reaction Q-value.

4.4 Why Inverse Reactions Matter in Stars

Inside stars, nuclear reactions occur within a thermal plasma. The nuclei possess a distribution of kinetic energies described approximately by the Maxwell–Boltzmann distribution. At low temperatures, most particles have kinetic energies much smaller than the reaction Q-value. Under these conditions, inverse reactions are negligible and the forward reaction dominates. However, as the temperature increases, the number of particles in the high-energy tail of the Maxwell–Boltzmann distribution increases significantly. Eventually, a fraction of the particles acquire enough energy to initiate the inverse reaction. The importance of inverse reactions therefore depends strongly on the ratio

$$\frac{Q}{kT}$$

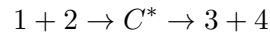
When $Q \gg kT$, inverse reactions are suppressed. When $Q \sim kT$ inverse reactions become significant. In explosive environments such as novae, X-ray bursts, and supernovae, inverse reactions may compete directly with the forward reaction and strongly influence the resulting nucleosynthesis. The net abundance change of a nucleus depends on the balance between the forward and inverse reaction rates rather than on either rate separately.

4.5 Compound Nucleus Description

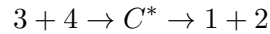
A powerful way to understand inverse reactions is through the concept of the compound nucleus. Suppose particles 1 and 2 collide and react. Rather than transforming immediately into the final products, the system often passes through an intermediate excited nucleus

$$C^*$$

The forward reaction may therefore be written as



The excited compound nucleus contains all the energy and *angular momentum*¹ of the incoming particles. After a short time, the compound nucleus decays into the final reaction products. The inverse reaction proceeds through exactly the same intermediate state,



The same compound nucleus is formed regardless of the direction in which the reaction proceeds. This observation is fundamental because it implies that the forward and inverse reactions are intimately connected. The existence of a common intermediate state forms the basis of inverse reaction theory

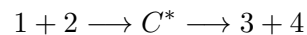


Figure 4.1: Forward and inverse reactions proceeding through the same compound nucleus.

4.6 Cross Section of a Nuclear Reaction

To relate forward and inverse reactions quantitatively, it is necessary to understand the structure of a nuclear reaction cross-section. The cross-section measures the probability that a reaction will occur when two particles interact. In quantum mechanics, the reaction probability depends on several factors

1. The wavelength associated with the incoming particles
2. The number of available quantum states
3. The symmetry properties of identical particles
4. The strength of the nuclear interaction

These factors combine to determine the magnitude of the reaction cross-section. Lets examine each of above contributions in the following sections

4.6.1 Quantum Mechanical Factor

According to wave mechanics, every particle possesses an associated de Broglie wavelength. For a relative momentum p

$$\lambda = \frac{h}{p}$$

Using the reduced wavelength

$$\bar{\lambda} = \frac{h}{p}$$

the geometrical contribution to the cross-section is proportional to

$$\pi \bar{\lambda}^2$$

Since

$$p = \sqrt{2\mu E}$$

¹Yes angular momentum but why ? think...

where μ is the reduced mass

$$\bar{\lambda} = \frac{\hbar}{\sqrt{2\mu E}}$$

Therefore,

$$\pi \bar{\lambda}^2 = \frac{\pi \hbar^2}{2\mu E}$$

This term represents the effective quantum-mechanical interaction area. Notice that the wavelength increases as the energy decreases. Consequently, low-energy particles possess larger interaction wavelengths and therefore larger geometrical reaction probabilities.

4.6.2 Statistical Factor

Angular momentum plays a central role in *all nuclear reactions*. Suppose the compound nucleus possesses total angular momentum J . Quantum mechanics allows

$$2J + 1$$

possible magnetic substates. If the entrance channel contains particles with spins J_1 and J_2 , the total number of available substates is

$$(2J_1 + 1)(2J_2 + 1)$$

The statistical factor is therefore defined as

$$\omega = \frac{2J + 1}{(2J_1 + 1)(2J_2 + 1)}$$

This factor measures the probability that the entrance channel couples to the compound nucleus state. The statistical factor is one of the key quantities appearing in the relationship between forward and inverse cross-sections

4.6.3 Kronecker Delta Correction

Special care must be taken when identical particles appear in the entrance or exit channel. For example,

$$p + p$$

contains two identical particles. Interchanging the particles does not produce a new physical state. To avoid double counting, a correction factor

$$\frac{1}{1 + \delta}$$

is introduced, where

$$\delta = \begin{cases} 1, & \text{identical particles} \\ 0, & \text{different particles} \end{cases}$$

This factor ensures the correct counting of quantum states.

4.6.4 Matrix Elements and Interaction Strength

The final ingredient in a nuclear cross-section is the interaction matrix element. Suppose the initial quantum state is denoted by

$$|i\rangle$$

and the final state by

$$|f\rangle$$

The probability amplitude for the transition is

$$\langle f|H|i\rangle$$

where H is the interaction Hamiltonian. The quantity

$$|\langle f|H|i\rangle|^2$$

determines the strength of the nuclear transition. A large matrix element corresponds to a highly probable reaction, while a small matrix element corresponds to a weak transition. The matrix element contains the detailed nuclear-structure information and represents the true dynamical content of the reaction. Combining all factors, the reaction cross-section may be written schematically as

$$\sigma \propto \pi \bar{\lambda}^2 \omega \frac{1}{1 + \delta} |\langle f|H|i\rangle|^2$$

This expression forms the starting point for deriving the reciprocity relations between forward and inverse nuclear Reactions

4.7 Time-Reversal Invariance

The connection between forward and inverse reactions is based on a fundamental principle of microscopic physics called **time-reversal invariance**. Consider the forward reaction

$$1 + 2 \rightarrow 3 + 4$$

In the compound nucleus picture, this reaction may be represented as

$$1 + 2 \rightarrow C^* \rightarrow 3 + 4$$

where C^* is an excited compound nucleus. The inverse reaction is

$$3 + 4 \rightarrow 1 + 2$$

which may similarly be written as

$$3 + 4 \rightarrow C^* \rightarrow 1 + 2$$

Thus, both the forward and inverse reactions proceed through the same compound nuclear state. The principle of time-reversal invariance states that the laws governing the microscopic nuclear process do not change when the direction of time is reversed. In other words, if the transition

$$1 + 2 \rightarrow C^* \rightarrow 3 + 4$$

is allowed, then the transition

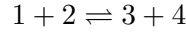
$$3 + 4 \rightarrow C^* \rightarrow 1 + 2$$

is also allowed, provided the required energy is available. This does not mean that the forward and inverse cross-sections are equal. The available phase space, spin degeneracies, reduced masses, and kinetic energies may differ in the two directions. However, the underlying nuclear matrix elements are related. For strong and electromagnetic nuclear reactions, no experimentally observed violation of time-reversal invariance is normally considered in standard nuclear astrophysics reaction-rate calculations. Therefore, the matrix elements governing the forward and inverse processes may be treated as equivalent in magnitude.

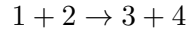
This principle allows the cross-section of an inverse reaction to be obtained from the cross-section of the corresponding forward reaction.

4.8 Reciprocity Theorem

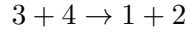
The reciprocity theorem gives the mathematical relation between the cross-section of a forward reaction and the cross-section of its inverse reaction. Consider the reaction pair



The forward reaction is



and the inverse reaction is



The cross-section for the forward reaction may be written schematically as

$$\sigma_{12} \propto \pi \bar{\lambda}_{12}^2 \omega_{12} (1 + \delta_{12}) |M_{12 \rightarrow 34}|^2$$

Similarly, the cross-section for the inverse reaction may be written as

$$\sigma_{34} \propto \pi \bar{\lambda}_{34}^2 \omega_{34} (1 + \delta_{34}) |M_{34 \rightarrow 12}|^2$$

Here,

$$\bar{\lambda}_{12} = \frac{\hbar}{\sqrt{2\mu_{12}E_{12}}}, \quad \bar{\lambda}_{34} = \frac{\hbar}{\sqrt{2\mu_{34}E_{34}}}$$

where

$$\mu_{12} = \frac{m_1 m_2}{m_1 + m_2} \quad \mu_{34} = \frac{m_3 m_4}{m_3 + m_4}$$

are the reduced masses in the entrance channels of the forward and inverse reactions. The quantities E_{12} and E_{34} denote the center-of-mass kinetic energies in the two channels. The statistical factors are

$$\omega_{12} = \frac{2J + 1}{(2J_1 + 1)(2J_2 + 1)},$$

and

$$\omega_{34} = \frac{2J + 1}{(2J_3 + 1)(2J_4 + 1)}$$

The same compound nucleus angular momentum J appears in both directions

4.8.1 Derivation of the Cross-Section Ratio

The forward cross-section may be written as

$$\sigma_{12} = \pi \bar{\lambda}_{12}^2 \frac{2J + 1}{(2J_1 + 1)(2J_2 + 1)} (1 + \delta_{12}) |\langle 3 + 4 | H_2 | C^* \rangle \langle C^* | H_1 | 1 + 2 \rangle|^2$$

The inverse cross-section is

$$\sigma_{34} = \pi \bar{\lambda}_{34}^2 \frac{2J + 1}{(2J_3 + 1)(2J_4 + 1)} (1 + \delta_{34}) |\langle 1 + 2 | H_1 | C^* \rangle \langle C^* | H_2 | 3 + 4 \rangle|^2$$

Here H_1 and H_2 represent the interaction operators associated with the two steps of the compound-nucleus process. Using time-reversal invariance, the matrix elements in the forward and inverse directions have the same magnitude. Therefore, when the ratio σ_{34}/σ_{12} is formed, the matrix-element factors cancel. Thus,

$$\frac{\sigma_{34}}{\sigma_{12}} = \frac{\bar{\lambda}_{34}^2 (2J_1 + 1)(2J_2 + 1)}{\bar{\lambda}_{12}^2 (2J_3 + 1)(2J_4 + 1)} \frac{1 + \delta_{34}}{1 + \delta_{12}}$$

Using

$$\bar{\lambda}^2 = \frac{\hbar^2}{2\mu E},$$

we obtain

$$\frac{\bar{\lambda}_{34}^2}{\bar{\lambda}_{12}^2} = \frac{\mu_{12} E_{12}}{\mu_{34} E_{34}}$$

Therefore,

$$\frac{\sigma_{34}}{\sigma_{12}} = \frac{\mu_{12} E_{12}}{\mu_{34} E_{34}} \frac{(2J_1 + 1)(2J_2 + 1)}{(2J_3 + 1)(2J_4 + 1)} \frac{1 + \delta_{34}}{1 + \delta_{12}}$$

This is the cross-section reciprocity relation. Equivalently, it may be written in the reverse form as

$$\sigma_{34} = \sigma_{12} \frac{\mu_{12} E_{12}}{\mu_{34} E_{34}} \frac{(2J_1 + 1)(2J_2 + 1)}{(2J_3 + 1)(2J_4 + 1)} \frac{1 + \delta_{34}}{1 + \delta_{12}}$$

This expression relates the inverse reaction cross-section to the forward reaction cross-section.

4.8.2 Physical Interpretation

The reciprocity theorem shows that the forward and inverse cross-sections are not independent quantities. The ratio depends on three main factors. First, it depends on the quantum mechanical wavelength factor

$$\frac{\mu_{12} E_{12}}{\mu_{34} E_{34}}$$

This factor arises because the effective interaction area is determined by the de Broglie wavelength of the particles in the entrance channel. Second, the ratio depends on the spin degeneracy factors,

$$\frac{(2J_1 + 1)(2J_2 + 1)}{(2J_3 + 1)(2J_4 + 1)}$$

These terms count the number of available quantum substates in the entrance channels of the forward and inverse reactions. Third, the ratio depends on the identical-particle correction,

$$\frac{1 + \delta_{34}}{1 + \delta_{12}}$$

This factor accounts for the correct counting of states when identical particles are present. The most important feature of the reciprocity theorem is that the detailed compound-nucleus matrix elements do not appear in the final ratio. They cancel because the same nuclear interaction governs both the forward and inverse processes. Therefore, measurement of the cross-section in one direction provides information about the cross-section in the reverse direction.

This is extremely useful in nuclear astrophysics because many reactions of astrophysical interest occur at energies that are difficult to access experimentally. In some cases, the inverse reaction is easier to measure in the laboratory, and the reciprocity theorem can then be used to infer the desired stellar reaction cross-section.

Chapter 5

Short Notes for Exam

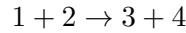
The reciprocity theorem relates cross-sections. However our stellar evolution and nucleosynthesis calculations require reaction rates. so natural next step is to derive the relationship between the forward and inverse thermonuclear reaction rates. The reaction rate per particle pair for two interacting particles is

$$\langle\sigma v\rangle = \left(\frac{8}{\pi\mu}\right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty E\sigma(E)e^{-E/kT} dE$$

This expression will be applied separately to the forward and inverse directions

5.0.1 Forward Reaction Rate

For the forward reaction



the reaction rate per particle pair is

$$\langle\sigma v\rangle_{12} = \left(\frac{8}{\pi\mu_{12}}\right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty E_{12}\sigma_{12}(E_{12})e^{-E_{12}/kT} dE_{12}$$

The total reaction rate per unit volume is

$$r_{12} = \frac{N_1 N_2}{1 + \delta_{12}} \langle\sigma v\rangle_{12}$$

The factor $1 + \delta_{12}$ avoids double counting when the two particles in the entrance channel are identical. For non-identical particles

$$\delta_{12} = 0$$

and therefore

$$r_{12} = N_1 N_2 \langle\sigma v\rangle_{12}.$$

For identical particles

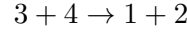
$$\delta_{12} = 1$$

and therefore

$$r_{12} = \frac{1}{2} N_1^2 \langle\sigma v\rangle_{12}$$

5.0.2 Inverse Reaction Rate

For the inverse reaction



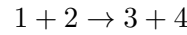
the reaction rate per particle pair is

$$\langle \sigma v \rangle_{34} = \left(\frac{8}{\pi \mu_{34}} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty E_{34} \sigma_{34}(E_{34}) e^{-E_{34}/kT} dE_{34}$$

The corresponding total inverse reaction rate per unit volume is

$$r_{34} = \frac{N_3 N_4}{1 + \delta_{34}} \langle \sigma v \rangle_{34}$$

The forward and inverse energies are related by the Q-value. For the forward reaction,



energy conservation gives

$$E_{34} = E_{12} + Q$$

Equivalently

$$E_{12} = E_{34} - Q$$

If $Q > 0$, the forward reaction releases energy. The inverse reaction then requires enough initial kinetic energy in the $3 + 4$ channel to overcome the energy deficit. Thus, inverse reactions are suppressed unless the particles in the inverse entrance channel possess sufficient thermal energy.

5.0.3 Ratio of Forward and Inverse Rates

The ratio of inverse to forward reaction rates can be obtained by combining the thermonuclear rate expressions with the cross-section reciprocity relation. After substituting the reciprocity relation into the inverse rate integral and using the relation between the forward and inverse energies, one obtains

$$\frac{\langle \sigma v \rangle_{34}}{\langle \sigma v \rangle_{12}} = \frac{(2J_1 + 1)(2J_2 + 1)}{(2J_3 + 1)(2J_4 + 1)} \frac{1 + \delta_{12}}{1 + \delta_{34}} \left(\frac{\mu_{12}}{\mu_{34}} \right)^{3/2} e^{-Q/kT}$$

This is the thermonuclear reciprocity relation for reaction rates. The most important factor is the exponential term,

$$e^{-Q/kT}$$

The remaining factors are often of order unity compared with the exponential temperature dependence. Thus, the approximate behavior is

$$\frac{\langle \sigma v \rangle_{34}}{\langle \sigma v \rangle_{12}} \propto e^{-Q/kT}$$

This result explains why inverse reactions are negligible at low temperatures but become increasingly important at high temperatures. If

$$kT \ll Q$$

then

$$e^{-Q/kT} \ll 1$$

and the inverse reaction rate is extremely small.

If

$$kT \sim Q$$

then the exponential suppression is reduced and inverse reactions may compete with the forward reaction. In terms of the temperature variable

$$T_9 = \frac{T}{10^9 \text{ K}}$$

one may write

$$\frac{Q}{kT} = \frac{11.605 Q}{T_9}$$

when Q is measured in MeV. Therefore,

$$\boxed{e^{-Q/kT} = e^{-11.605Q/T_9}}$$

This form is particularly useful in nuclear astrophysics because stellar temperatures are often expressed in gigakelvin and nuclear Q -values in MeV. For example if $Q = 1$ MeV, then at $T_9 = 1$

$$e^{-Q/kT} = e^{-11.605} \approx 9.1 \times 10^{-6}$$

Thus, even at 10^9 K, an inverse reaction with $Q = 1$ MeV is still strongly suppressed. At higher temperatures, however, the suppression becomes much weaker. This is why inverse reactions are especially important in explosive astrophysical environments such as novae, supernovae, and X-ray bursts. The ratio of the total reaction rates is

$$\frac{r_{34}}{r_{12}} = \frac{N_3 N_4}{N_1 N_2} \frac{1 + \delta_{12}}{1 + \delta_{34}} \frac{\langle \sigma v \rangle_{34}}{\langle \sigma v \rangle_{12}}$$

Substituting the thermonuclear reciprocity relation gives

$$\boxed{\frac{r_{34}}{r_{12}} = \frac{N_3 N_4}{N_1 N_2} \frac{(2J_1 + 1)(2J_2 + 1)}{(2J_3 + 1)(2J_4 + 1)} \left(\frac{\mu_{12}}{\mu_{34}} \right)^{3/2} e^{-Q/kT}}$$

This expression shows explicitly that the competition between forward and inverse reactions depends on abundance, spin degeneracy, reduced masses, Q -value and temperature. The exponential factor remains the dominant temperature-sensitive term.

5.1 Temperature Dependence

One of the most important consequences of inverse-reaction theory is the strong dependence of inverse reaction rates on temperature. The ratio of inverse to forward reaction rates was shown to be

$$\frac{\langle \sigma v \rangle_{34}}{\langle \sigma v \rangle_{12}} = C e^{-Q/kT}$$

where C contains statistical and mass-dependent factors. The dominant temperature dependence is therefore determined by the exponential term

$$e^{-Q/kT}$$

This factor controls the probability that particles in a thermal plasma possess sufficient energy to initiate the inverse reaction.

5.1.1 Role of the Exponential Factor

The Maxwell–Boltzmann distribution contains the factor

$$e^{-E/kT}$$

As the energy increases, the number of particles possessing that energy decreases exponentially. Inverse reactions require particles with energies above the threshold energy. Consequently, only particles located in the extreme high-energy tail of the Maxwell Boltzmann distribution can participate. The probability of finding such particles is approximately

$$e^{-Q/kT}$$

where Q is the reaction Q-value. Thus, inverse reactions are exponentially suppressed relative to the forward reaction.

5.1.2 Dependence on the Q-Value

The suppression of inverse reactions depends directly on the reaction Q-value. Large-Q reactions require a larger energy input for the inverse process. Consequently

$$Q \uparrow \implies e^{-Q/kT} \downarrow$$

Therefore, reactions with large positive Q-values have strongly suppressed inverse rates. Small-Q reactions exhibit significantly weaker suppression and may achieve near-equilibrium conditions at comparatively low temperatures.

5.1.3 Low-Temperature Limit

At low temperatures

$$kT \ll Q$$

In this limit

$$e^{-Q/kT} \rightarrow 0$$

Consequently

$$\langle \sigma v \rangle_{34} \ll \langle \sigma v \rangle_{12}$$

Only the forward reaction contributes significantly. Most hydrogen burning and helium burning reactions in ordinary stars operate in this regime. Under these conditions inverse reactions may often be neglected.

5.1.4 High-Temperature Limit

At sufficiently high temperatures,

$$kT \sim Q$$

The exponential suppression becomes much weaker. As a result,

$$\langle \sigma v \rangle_{34}$$

can become comparable to

$$\langle \sigma v \rangle_{12}$$

Both reaction directions must then be included in the reaction network. Such conditions occur in

- novae
- X-ray bursts
- supernova explosions
- nuclear statistical equilibrium

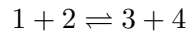
In these environments, inverse reactions strongly influence the abundance distribution.

5.2 Detailed Balance

When both forward and inverse reactions occur simultaneously, the system may approach equilibrium. The principle governing this equilibrium is known as **detailed balance**. Detailed balance provides the fundamental connection between nuclear reaction rates and thermodynamic equilibrium.

5.2.1 Thermodynamic Equilibrium

A system is said to be in thermodynamic equilibrium when all macroscopic properties remain constant with time. In equilibrium, the rate at which a reaction produces a nucleus must equal the rate at which that nucleus is destroyed. For the reaction



equilibrium requires

$$r_{12} = r_{34}$$

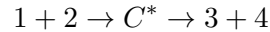
Therefore,

$$\frac{dN_i}{dt} = 0$$

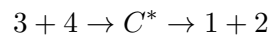
for all participating nuclei

5.2.2 Microscopic Reversibility

Detailed balance originates from the principle of microscopic reversibility. Microscopic reversibility states that every elementary process occurring in one direction is accompanied by the corresponding reverse process. For example,



must be accompanied by



At equilibrium, the number of transitions occurring in each direction is exactly the same. Thus,

$$r_{12} = r_{34}$$

This equality forms the foundation of inverse-reaction theory

5.2.3 Connection to Stellar Equilibrium

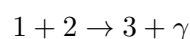
Many stellar environments do not achieve complete thermodynamic equilibrium. However local equilibrium may occur within limited groups of nuclei. Examples include

- quasi-statistical equilibrium
- nuclear statistical equilibrium
- equilibrium between radiative capture and photodisintegration

The principle of detailed balance allows inverse reaction rates to be calculated from experimentally measured forward reaction rates.

5.3 Radiative Capture Reactions

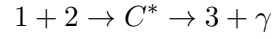
Radiative capture reactions are among the most important reactions in stellar nucleosynthesis. A radiative capture reaction has the form



The reaction proceeds through the formation of a compound nucleus followed by gamma-ray emission

5.3.1 Formation of Heavy Nuclei

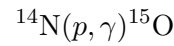
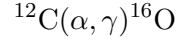
When two nuclei combine, the resulting nucleus is usually produced in an excited state. The excess energy must be removed in order for the final nucleus to remain bound. This energy is carried away by a gamma ray. Thus,



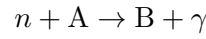
Radiative capture therefore provides the primary mechanism by which light nuclei are assembled into heavier nuclei.

5.3.2 Examples

Important examples include



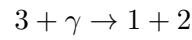
and



These reactions play essential roles in stellar nucleosynthesis.

5.4 Photodisintegration Reactions

The inverse of radiative capture is photodisintegration. A gamma ray interacts with a nucleus and causes it to break apart.



Photodisintegration reactions become important at high temperatures where energetic gamma rays are abundant

5.4.1 Threshold Energy

A gamma ray can induce a reaction only if its energy exceeds the threshold energy. The threshold energy is approximately equal to the Q-value of the inverse reaction. Therefore,

$$E_\gamma \geq E_{\text{thr}}$$

Gamma rays below this energy cannot induce photodisintegration

5.4.2 Planck Distribution

The energy distribution of photons in thermal equilibrium is described by Planck's law. The photon number density is

$$N_\gamma(E_\gamma)dE_\gamma = \frac{8\pi}{(hc)^3} \frac{E_\gamma^2}{e^{E_\gamma/kT} - 1} dE_\gamma$$

This distribution replaces the Maxwell–Boltzmann distribution used for massive particles

5.4.3 Photodisintegration Rate

The gamma-induced decay constant is

$$\lambda_\gamma = \int_0^\infty c N_\gamma(E_\gamma) \sigma(E_\gamma) dE_\gamma$$

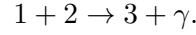
Substituting Planck's distribution yields

$$\lambda_\gamma = \frac{8\pi}{h^3 c^2} \int_{E_{\text{thr}}}^\infty \frac{E_\gamma^2 \sigma(E_\gamma)}{e^{E_\gamma/kT} - 1} dE_\gamma$$

5.5 Forward and Inverse Reactions Involving Gamma Rays

5.5.1 Radiative Capture

The forward reaction is

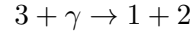


The reaction rate is

$$r_{12} = N_1 N_2 \langle \sigma v \rangle_{12}$$

5.5.2 Photodisintegration

The inverse reaction is



The reaction rate becomes

$$r_{\gamma 3} = N_3 \lambda_\gamma$$

5.5.3 Reaction Rate Ratio

Using detailed balance and reciprocity relations, one obtains

$$\frac{\lambda_\gamma}{\langle \sigma v \rangle} \propto T^{3/2} e^{-Q/kT}$$

The exponential factor again controls the temperature dependence

5.6 Net Stellar Reaction Rate

In stellar matter, both forward and inverse reactions occur simultaneously. The net reaction rate is therefore

$$r_{\text{net}} = r_{\text{forward}} - r_{\text{inverse}}$$

For a two-body reaction

$$r_{\text{net}} = \frac{N_1 N_2}{1 + \delta_{12}} \langle \sigma v \rangle_{12} - \frac{N_3 N_4}{1 + \delta_{34}} \langle \sigma v \rangle_{34}$$

Only the net rate changes the abundance distribution

5.7 Energy Generation in Stars

The forward reaction releases energy

$$Q r_{12}$$

The inverse reaction absorbs energy $Q r_{34}$. The net energy generation rate is therefore

$$\epsilon = Q(r_{12} - r_{34})$$

Inverse reactions reduce the net energy released by stellar nuclear burning

5.8 Influence on Stellar Evolution

Inverse reactions become increasingly important as temperature increases. They influence

- reaction flow
- isotopic abundances
- stellar energy production
- nucleosynthesis pathways
- equilibrium abundance distributions

In extreme environments, inverse reactions can dominate the nuclear reaction network.

5.9 Applications

5.9.1 Hydrogen Burning

Inverse reactions are negligible because

$$kT \ll Q$$

The reaction flow proceeds almost entirely in the forward direction.

5.9.2 Helium Burning

Most inverse reactions remain weak, although some become significant in the late stages of stellar evolution.

5.9.3 CNO Cycle

Inverse reactions slightly modify equilibrium abundances but generally do not dominate the reaction network.

5.9.4 Explosive Nucleosynthesis

Temperatures become sufficiently high that inverse reactions strongly compete with forward reactions. Photodisintegration reactions become particularly important.

5.9.5 Nuclear Statistical Equilibrium

At temperatures exceeding several GK,

$$r_{\text{forward}} = r_{\text{inverse}}.$$

The abundance distribution is determined by statistical equilibrium rather than individual reaction rates.

5.10 Worked Examples

Consider a reaction with

$$Q = 2 \text{ MeV}.$$

At

$$T_9 = 1,$$

the suppression factor is

$$e^{-11.605Q/T_9} = e^{-23.21} \approx 8.3 \times 10^{-11}.$$

Thus, the inverse reaction is negligible.

At

$$T_9 = 5,$$

the factor becomes

$$e^{-4.64} \approx 9.7 \times 10^{-3}.$$

The inverse reaction is still suppressed but becomes increasingly important.

5.11 Summary

Inverse reactions are governed by the reciprocity theorem, detailed balance, and thermodynamic equilibrium.

Their importance depends primarily on the exponential factor

$$e^{-Q/kT}.$$

At low temperatures, inverse reactions are negligible.

At high temperatures, inverse reactions may compete with or even dominate forward reactions.

The balance between forward and inverse reactions determines stellar abundances, energy generation, and the evolution of astrophysical reaction networks.